

Neil Guertin

Software Quality Assurance Analyst
Software Test Automation Developer
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EXPERIENCE	Collabora Productivity					Remote
	Software Engineer in Test					Dec. 2023 – Jul. 2024
	- Supported development of a web application for collaborative document editing - Performed all automated testing activities as the only test engineer in a small company - Reduced testing time by 50%, increased reliability of tests, and improved failure diagnostics - Created performance testing framework and systems for monitoring test failures					
	MathWorks					Natick, MA
	Quality Engineer <i>MATLAB Language Front End</i>					Jun. 2018 – Jun. 2022
	- Wrote tests and testing tools for MATLAB language and syntax, including individual tests for bug fixes and complete suites with documentation for new features - Maintained existing legacy test bed, and updated tests to meet modern standards - Found and investigated bugs in MATLAB language and syntax - Provided feedback on testability and usability of new features					
	Application Support Engineer <i>Engineering Development Group</i>					Jun. 2017 – Jun. 2018
	- Provided technical support to users of MATLAB - Documented and shared solutions to assist with future cases - Improved automated crash reporting and response system					
	Quality Engineer Intern <i>MATLAB Graphics & Charting</i>					Jun. 2016 – Dec. 2016
	- Wrote testing tools for MATLAB charts including mixins and a comparator - Updated and improved a large collection of legacy tests					
EDUCATION	Rochester Institute of Technology					Rochester, NY
	Lab Programming Researcher <i>Center for Detectors</i>					Jun. 2015 – Dec. 2015
	- Assisted faculty and engineers on NASA and NSF funded research - Wrote IDL code to gather, reduce, analyze, plot, and simulate data - Improved temperature and pressure logging capabilities on four cryogenic dewar systems					
	Student Lab Instructor/TA <i>Computer Science Department</i>					Sep. 2013 – May 2017
	- Graded homeworks and labs using custom scripts - Tutored students in class, in the mentoring center, and privately					
	Rochester Institute of Technology					Rochester, NY
	Bachelor of Science <i>Computer Science & Computational Mathematics</i>					Sep. 2012 – May 2017
	- College of Science Deans List x7 - Magna Cum Laude - GPA 3.8					
SKILLS	C/C++ JavaScript Cypress/Mocha Linux Algorithms FLOSS Development					
	MATLAB TypeScript Automated Tests Git/GitHub Data Structures Agile Development					
	Java Angular Performance Tests Self-Hosting Object-Oriented Programming Test-Driven Development					
	Python PHP Debugging Docker Programming Languages Technical Support					
	Bash HTML/CSS Bug Triage Full Stack Compilers Remote Work					
	MariaDB/MySQL Jenkins/CI/CD					
LANGUAGES	English Fluent					
	French Proficient					

OTHER ACTIVITIES

ContraMontreal

Montreal, QC

Organizer

2022 – Present

- Organize more than 25 traditional social dance events per year, open to the general public
- Plan work and solve problems on a team of about ten volunteers

Webmaster

2023 – Present

- Build and manage website
- Administer email account, mailing lists, and other online accounts

Sound Technician

2024 – Present

- Set up sound equipment and monitor levels and EQ during each event

Treasurer

2025 – Present

- Track and report financial transactions for the organization

OpenStreetMap

Contributor

2022 – Present

- Contribute to open geographic data usable by anyone
- Write wiki pages to help other contributors and standardize tags

PROJECTS

NEFFA Volunteer Scheduler Website *New England Folk Festival Association* Summer 2026

PHP, HTML, MariaDB/MySQL

- Clean up and fix legacy LAMP project used by a folk organization to schedule volunteers
- Add Cypress tests to ensure website continues to work into the future
- Ensure compatibility with 20 years of existing data

ContraMontreal Website <https://www.contramontreal.org> *ContraMontreal*

Fall 2024

Jekyll, HTML, CSS, JavaScript

- Designed and built static website in Jekyll to replace an old Wordpress site
- Balanced technical needs of website with need for non-technical members to make updates
- Generated .ics files for upcoming events for use with Webcal
- Scrutinized details such as favicons, fonts, image sizes, users without JavaScript, and sitemaps

My Timeline *Personal*

2022 – Present

Java, Angular, Data Processing

- Gathered, processed, and merged location data from many different sources
- Generated csv of minute-by-minute location annotated with travel information and other labels
- Built web application in Angular to view and explore the data
- Created and styled maps in QGIS

Footsteps *Personal*

2023 – Present

Angular, Python, Full Stack

- Full stack application for viewing running activity data built with Angular and Python
- Focus on data visualization including maps, graphs, charts, and summaries

P-Code R2022a Format *MathWorks*

2020 – 2022

MATLAB, Test Design, Project Management

- Led testing activities for important multi-year project
- Wrote test plans, implemented tests, found and reported bugs
- Qualified and delivered betas to select customers
- Coordinated with other QEs on impacted downstream teams across the company

AST Test Data *MathWorks*

Winter 2021

MATLAB, Programming Languages, Test Design

- Wrote data set of MATLAB files representing all Abstract Syntax Tree nodes and cases
- Added tests for downstream clients of the AST, finding dozens of issues
- Saved development time for new language features while increasing confidence in quality

GraphicsComparator *MathWorks*

Fall 2016

MATLAB, Test Design

- Built a powerful and flexible comparator for graphics objects in MATLABUnit framework
- Wrote tests and found bugs in save/load, copy/paste, and other related workflows
- Provided feedback to MATLABUnit on the usability of their comparator framework